

Sadeep Withana

Computer Engineering Undergraduate

Embedded Systems · IoT · TinyML · Edge AI · Robotics

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PROFILE

Computer Engineering undergraduate at General Sir John Kotelawala Defence University (KDU) who builds hardware-software systems: sensors, microcontrollers, embedded machine learning and connected devices. Currently researching how INT8 quantization affects a compact ECG model for atrial fibrillation detection on the ESP32-S3. Interested in embedded systems, IoT, TinyML, edge AI, robotics, and cloud/DevOps engineering roles and research opportunities.

EDUCATION

BSc (Hons) in Computer Engineering

In progress

General Sir John Kotelawala Defence University (KDU)

Faculty of Computing · Intake 41

Relevant coursework: Embedded Systems, Internet of Things, Digital Signal Processing, Computer Network Security, Big Data Analytics, Computer Systems Engineering, Engineering Design Project

RESEARCH

Impact of INT8 Quantization on Compact ECG-Based AF Detection on ESP32-S3

Ongoing

- Research question: how INT8 quantization affects AF detection performance, memory usage and inference time of a compact 1D CNN on a microcontroller.
- Pipeline: single-lead ECG dataset, preprocessing, Float32 baseline, post-training INT8 quantization, deployment with TensorFlow Lite Micro on ESP32-S3.
- Evaluation: detection performance, memory, inference time, model size and compute requirements.

PROJECTS

IoT Fall Detection System

In development

ESP32 · MPU6050 · Random Forest / SVM · KiCad · OpenSCAD

- Fall detection node for elderly care using IMU data and embedded ML classifiers, with post-fall inactivity analysis to reduce false alarms.
- Designed the PCB in KiCad and a parametric 3D-printed enclosure in OpenSCAD.

WellNest — Privacy-First Ambient Assisted Living Multi-Agent System

Team prototype

FastAPI · Google ADK · Gemini · A2A · MCP · FHIR R4 · ESP32-S3

- Camera-free elderly care monitoring with four cooperating agents on an asynchronous pub/sub event bus.
- Team entry for AI Challenge Sri Lanka 2026 (Phase II). My role: backend design and UI/UX.

Garden Watering Robot

In progress

ESP32-S3 · IR line sensors · Motor driver · Soil moisture sensors

- Line-following robot that visits two garden zones, reads soil moisture and waters only the zones that need it.

Face Mask Detection

Built

Python · OpenCV · MobileNetV2 · Flask

- Camera → face detection → preprocessing → MobileNetV2 classifier → mask / no-mask label, delivered through a Flask web app.

IoT & Sensor Builds

Ongoing

ESP32 · ESP8266 · Arduino · MQTT · ThingsBoard · ThingSpeak

- Sensor and actuator builds (DHT, MQ, ultrasonic, servo, relays, water pumps) publishing to cloud dashboards over MQTT.

SKILLS

Comfortable with

Python · C/C++ · Arduino · ESP32 · Git/GitHub · Basic IoT development

Currently exploring

TinyML · Edge AI · TensorFlow Lite Micro · ESP32-S3 · Model quantization · Robotics · DevOps

Hardware & tools

KiCad · EasyEDA · OpenSCAD · 3D printing · Linux · MQTT · ThingsBoard · ThingSpeak

RESEARCH INTERESTS

Embedded AI · Biomedical IoT · ECG analysis · Edge computing · Resource-constrained ML